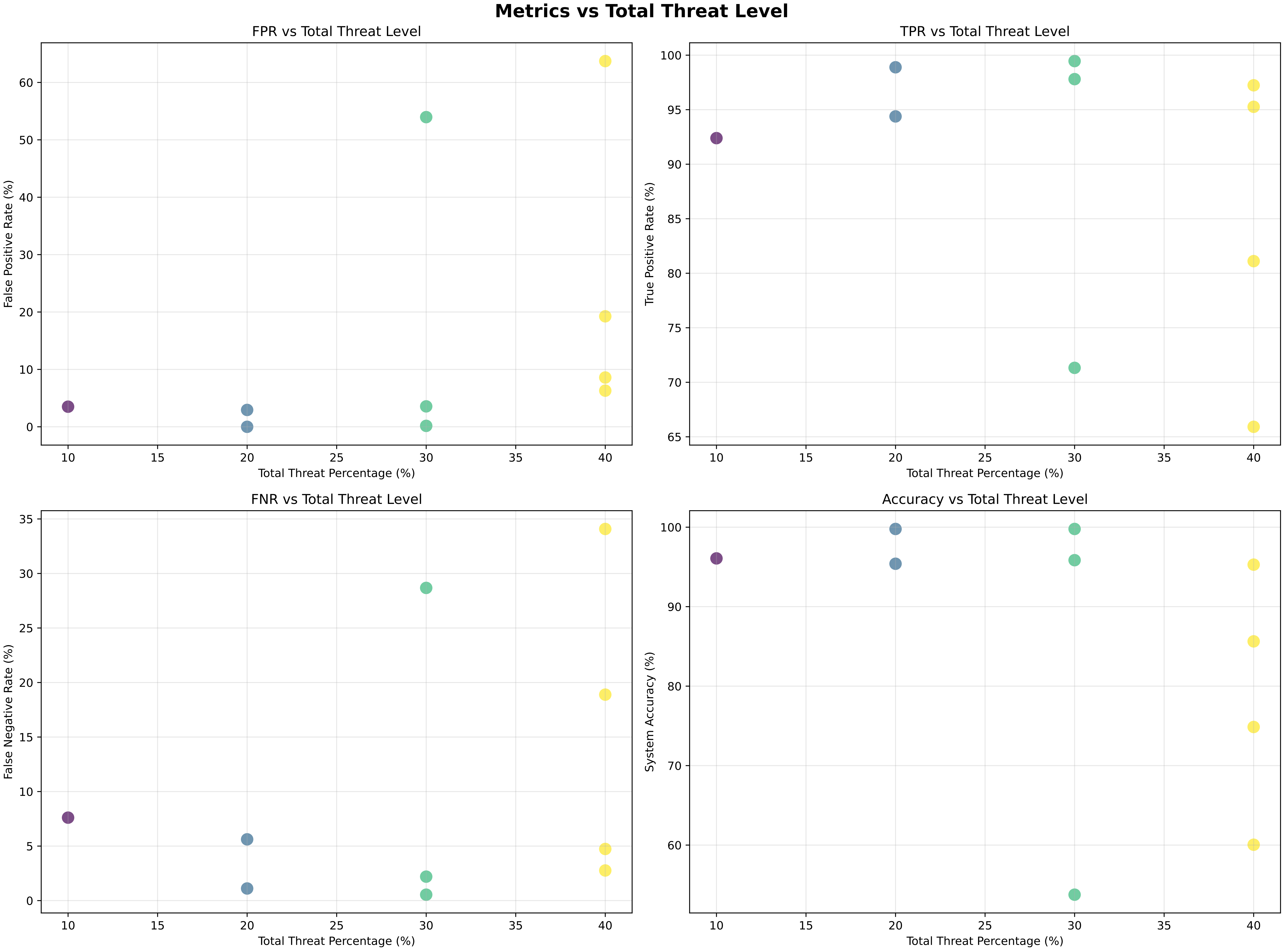
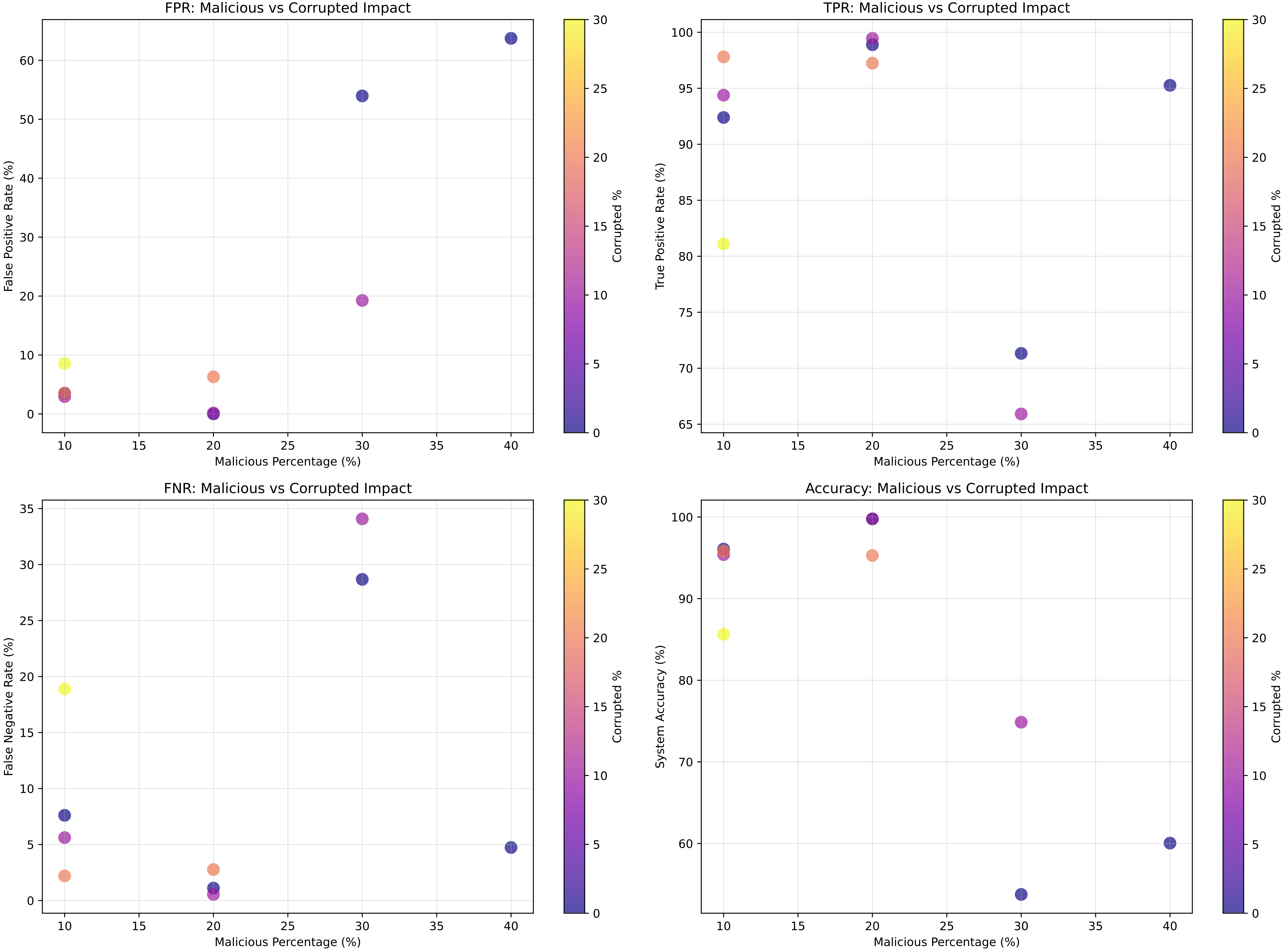
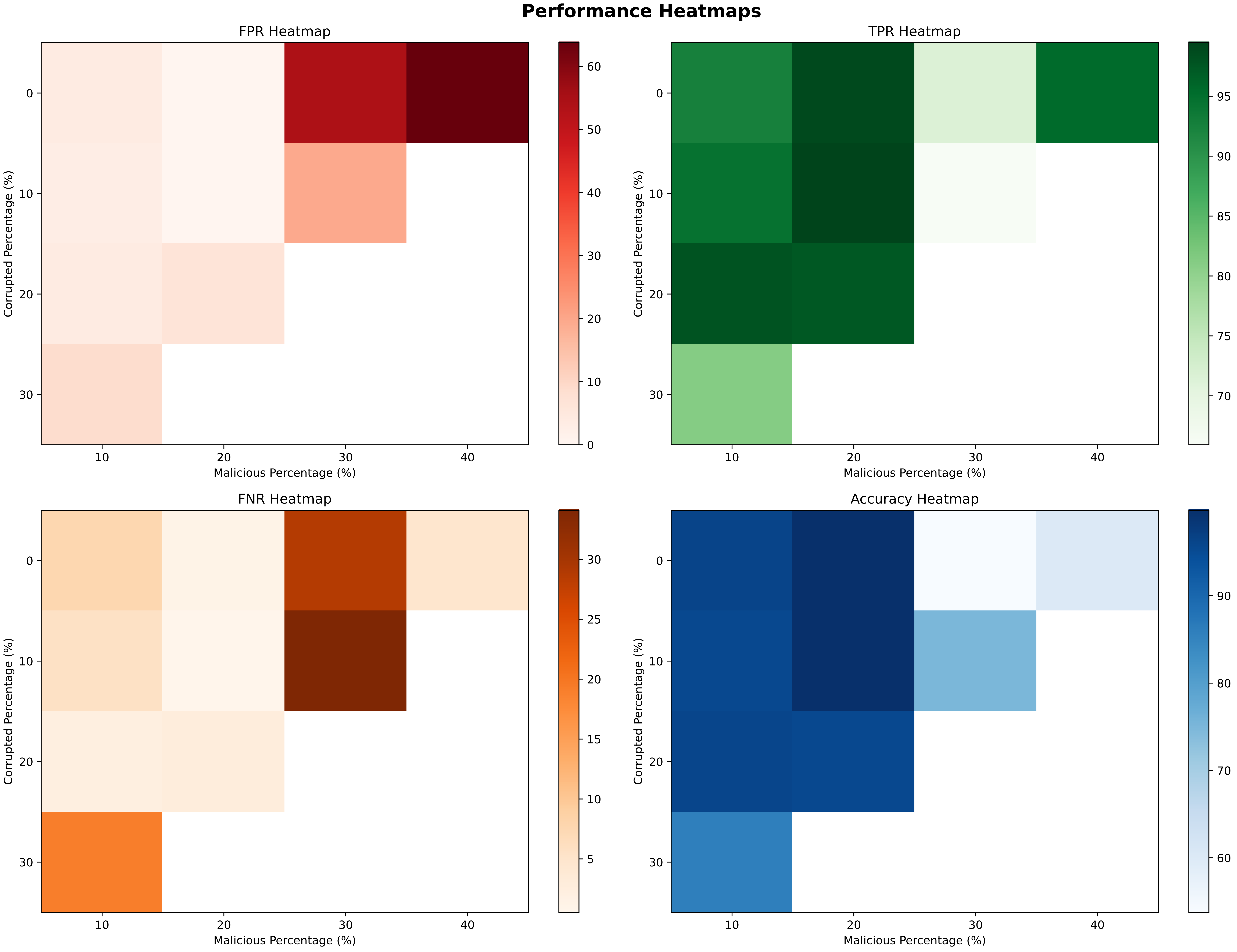

Metrics vs Total Threat Level
FPR vs Total Threat Level
False Positive Rate (%)
0
10
20
30
40
50
60
Total Threat Percentage (%)
10
15
20
25
30
35
40
TPR vs Total Threat Level
True Positive Rate (%)
65
70
75
80
85
90
95
100
Total Threat Percentage (%)
10
15
20
25
30
35
40
FNR vs Total Threat Level
False Negative Rate (%)
0
5
10
15
20
25
30
35
Total Threat Percentage (%)
10
15
20
25
30
35
40
Accuracy vs Total Threat Level
System Accuracy (%)
60
70
80
90
100
Total Threat Percentage (%)
10
15
20
25
30
35
40



Malicious vs Corrupted Percentage Impact


Performance Heatmaps
FPR Heatmap
TPR Heatmap
FNR Heatmap
Accuracy Heatmap
Corrupted Percentage (%)
Malicious Percentage (%)
0
10
20
30
40
60
50
40
30
20
10
0
95
90
85
80
75
70
25
15
5
60

Configuration Effectiveness Ranking

```
PERCENTAGE CONFIGURATION RANKING
========================================

1. 20%M, 10%C (Total: 30%)
   FPR: 0.2%, TPR: 99.5%, FNR: 0.5%, Acc: 99.8%
   Score: 198.5

2. 20%M, 0%C (Total: 20%)
   FPR: 0.0%, TPR: 98.9%, FNR: 1.1%, Acc: 99.8%
   Score: 197.6

3. 10%M, 20%C (Total: 30%)
   FPR: 3.6%, TPR: 97.8%, FNR: 2.2%, Acc: 95.8%
   Score: 187.9

4. 20%M, 20%C (Total: 40%)
   FPR: 6.3%, TPR: 97.2%, FNR: 2.8%, Acc: 95.3%
   Score: 183.5

5. 10%M, 10%C (Total: 20%)
   FPR: 2.9%, TPR: 94.4%, FNR: 5.6%, Acc: 95.4%
   Score: 181.2

6. 10%M, 0%C (Total: 10%)
   FPR: 3.5%, TPR: 92.4%, FNR: 7.6%, Acc: 96.1%
   Score: 177.4

7. 10%M, 30%C (Total: 40%)
   FPR: 8.6%, TPR: 81.1%, FNR: 18.9%, Acc: 85.6%
   Score: 139.3

8. 30%M, 10%C (Total: 40%)
   FPR: 19.3%, TPR: 65.9%, FNR: 34.1%, Acc: 74.9%
   Score: 87.4

9. 40%M, 0%C (Total: 40%)
   FPR: 63.7%, TPR: 95.3%, FNR: 4.7%, Acc: 60.0%
   Score: 86.9

10. 30%M, 0%C (Total: 30%)
   FPR: 54.0%, TPR: 71.3%, FNR: 28.7%, Acc: 53.8%
   Score: 42.4


PERCENTAGE TESTING:
This analysis tests different combinations of malicious
and corrupted peer percentages to understand how
threat levels affect system performance.

CONFIGURATIONS TESTED:
Total configurations: 10
Total threat levels: 10% to 40%
```

Summary Statistics

```
PERCENTAGE COMPARISON ANALYSIS - SUMMARY
========================================

Generated: 2025-09-01 13:00:43
Results Path: FL/results_percentages

CONFIGURATIONS ANALYZED:
• Total configurations: 10
• Malicious percentages: [np.int64(10), np.int64(20), np.int64(30), np.int64(40)]
• Corrupted percentages: [np.int64(0), np.int64(10), np.int64(20), np.int64(30)]
• Total threat levels: [np.int64(10), np.int64(20), np.int64(30), np.int64(40)]

DETAILED RESULTS:
\n• 10%M, 0%C (Total: 10%):\n  FPR=3.5%, TPR=92.4%, FNR=7.6%, Acc=96.1%\n\n• 10%M, 10%C (Total: 20%):\n  FPR=2.9%, TPR=94.4%, FNR=5.6%, Acc=95.4%\n\n• 10%M, 20%C (Total: 30%):\n  FPR=3.6%, TPR=97.8%, FNR=2.2%, Acc=95.8%\n\n• 10%M, 30%C (Total: 40%):\n  FPR=8.6%, TPR=81.1%, FNR=18.9%, Acc=85.6%\n\n• 20%M, 0%C (Total: 20%):\n  FPR=0.0%, TPR=98.9%, FNR=1.1%, Acc=99.8%\n\n• 20%M, 10%C (Total: 30%):\n  FPR=0.2%, TPR=99.5%, FNR=0.5%, Acc=99.8%\n\n• 20%M, 20%C (Total: 40%):\n  FPR=6.3%, TPR=97.2%, FNR=2.8%, Acc=95.3%\n\n• 30%M, 0%C (Total: 30%):\n  FPR=54.0%, TPR=71.3%, FNR=28.7%, Acc=53.8%\n\n• 30%M, 10%C (Total: 40%):\n  FPR=19.3%, TPR=65.9%, FNR=34.1%, Acc=74.9%\n\n• 40%M, 0%C (Total: 40%):\n  FPR=63.7%, TPR=95.3%, FNR=4.7%, Acc=60.0%\n

METRICS EXPLANATION:
• FPR (False Positive Rate): Good nodes incorrectly removed / Total good nodes sent
• TPR (True Positive Rate): Malicious nodes correctly detected / Total malicious nodes sent
• FNR (False Negative Rate): Malicious nodes incorrectly confirmed / Total malicious nodes sent
• Accuracy: Overall system accuracy (correctly handled nodes / total nodes)

ANALYSIS INSIGHTS:
• Total threat level ranges from 10% to 40%
• Different malicious/corrupted ratios at same total threat level
• System performance under varying adversarial conditions
• Optimal balance between security and false positives

PERCENTAGE PARAMETERS:
The malicious_percentage and corrupted_percentage parameters control
the proportion of adversarial peers in the federated learning system:
• Malicious peers: Actively try to degrade model performance
• Corrupted peers: Have degraded data quality but not malicious intent
• Total threat = malicious% + corrupted%
• Higher percentages create more challenging environments
```